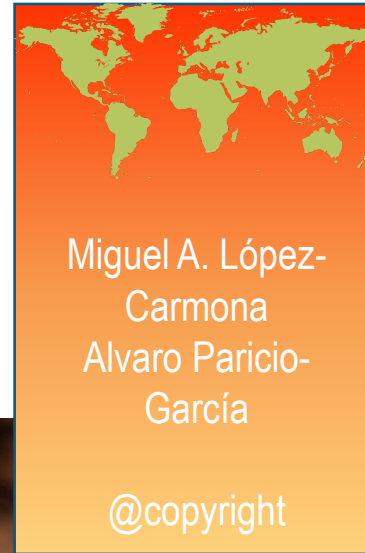


Digital Twin Practice

Advanced simulation
strategies for
managing risks

Variation experiments:

- Demand
- Transportation cost
- and others



European University Alliance for Global Health



Co-funded by the
European Union



Universidad
de Alcalá



Learning objectives

1. Develop analytical and management skills to analyze the effect of random variations in the supply chain.
2. Develop risk analysis and management skills.
 1. To understand the relevance of randomized experiments.
 2. To be able to simulate probabilistic environments.



What are variation experiments?

The screenshot shows the anyLogistix 3 PLE interface. The 'Experiment settings' dialog box is open, displaying various configuration options. The 'Variation experiment' option is highlighted with a red dashed box. The dialog box includes sections for 'Experiment start', 'Experiment end', 'Use replications', 'Replications per iteration', 'Number of threads to use', 'Variable parameters', 'Finances stats unit', 'Product stats unit', 'Time stats unit', and 'Distance stats unit'.

- So far we have been "forcing" situations on a case-by-case basis to analyze the effects on the supply chain.
- It is normal that several situations occur simultaneously and that we have a lot of combinations to simulate.
- Example:
 - 10 customers
 - 10 possible demand values
 - 5 possible cost ranges
 - 10 possible min-max inventory ranges→ There would be $10 \times 10 \times 5 \times 10 \times 10 = 5,000$ possible experiments.
- Anylogistix can be configured to generate variations on the total possible combinations.



What can be varied?

- BOM - Bill of Materials
- Customers
- DC and factories
- Demand
- Events
- Inventory
- Locations
- Routes
- Periods
- Production
- Products
- Supply
- Suppliers



We continue with WHC-Mobile



As you have proven to be good supply chain analysts, I will continue to rely on you for further analysis.



Supplier Taiwan, Supplier China



Factory



DC



(All customers)





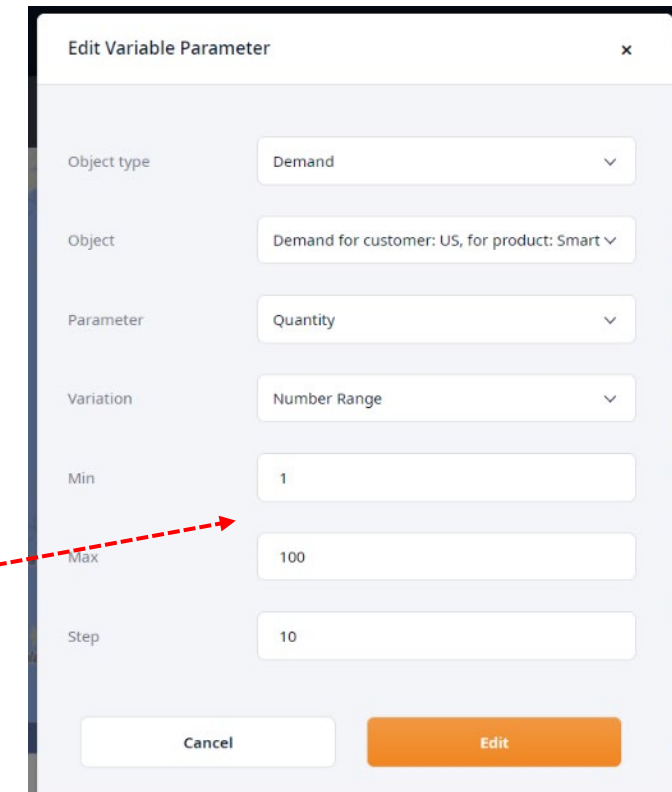
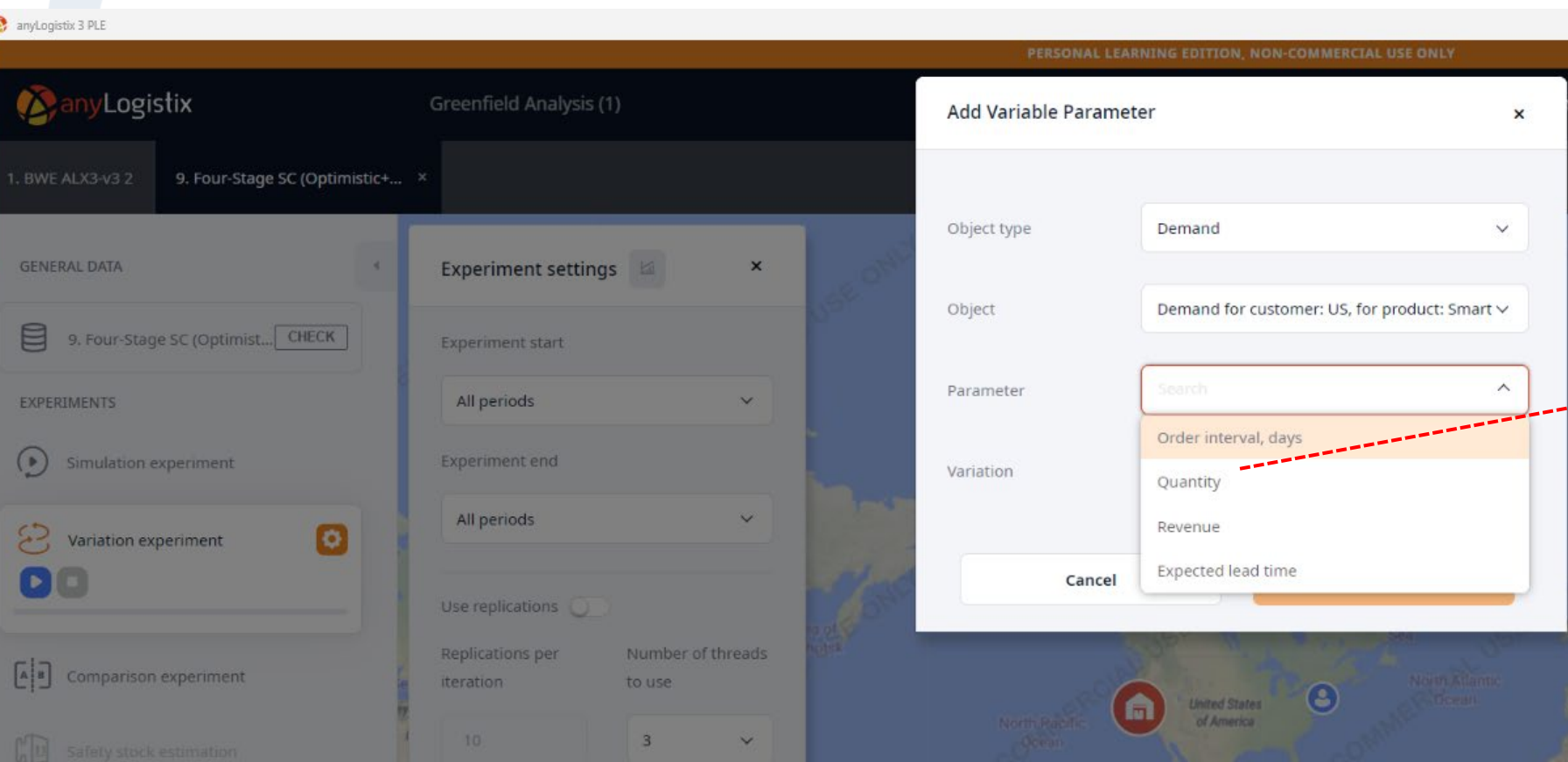
Exercise 10: Demand variation.

- Demand variation
- Best-Mean-Worst análisis
- Variation experiments



Demand variation scenario

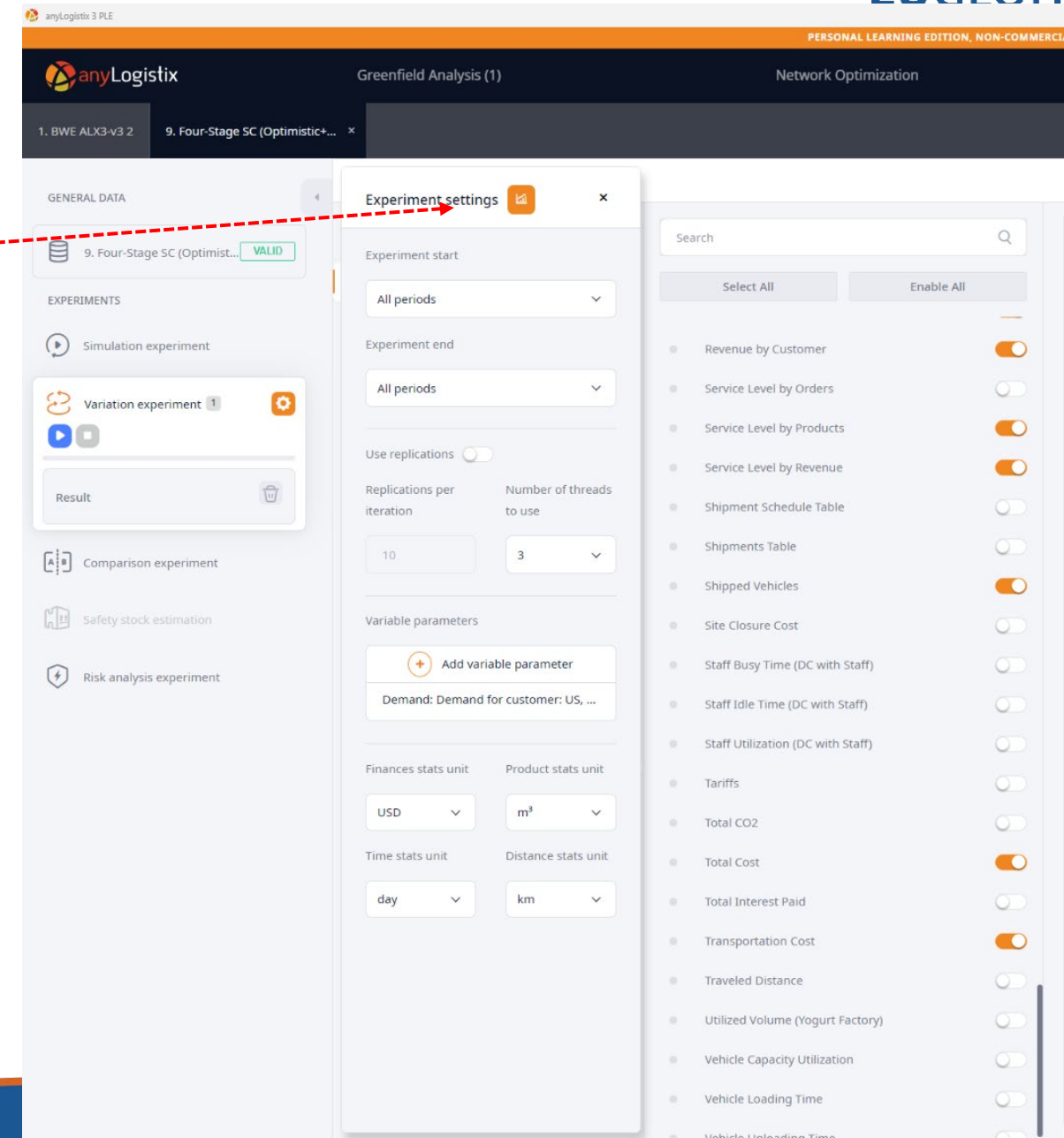
- Example: varying the demand of US customers from the current value of 10 to values between [1,100] in steps of 10.





Demand variation scenario

- In order to make a good simulation, we must also select the KPIs we want to analyze.

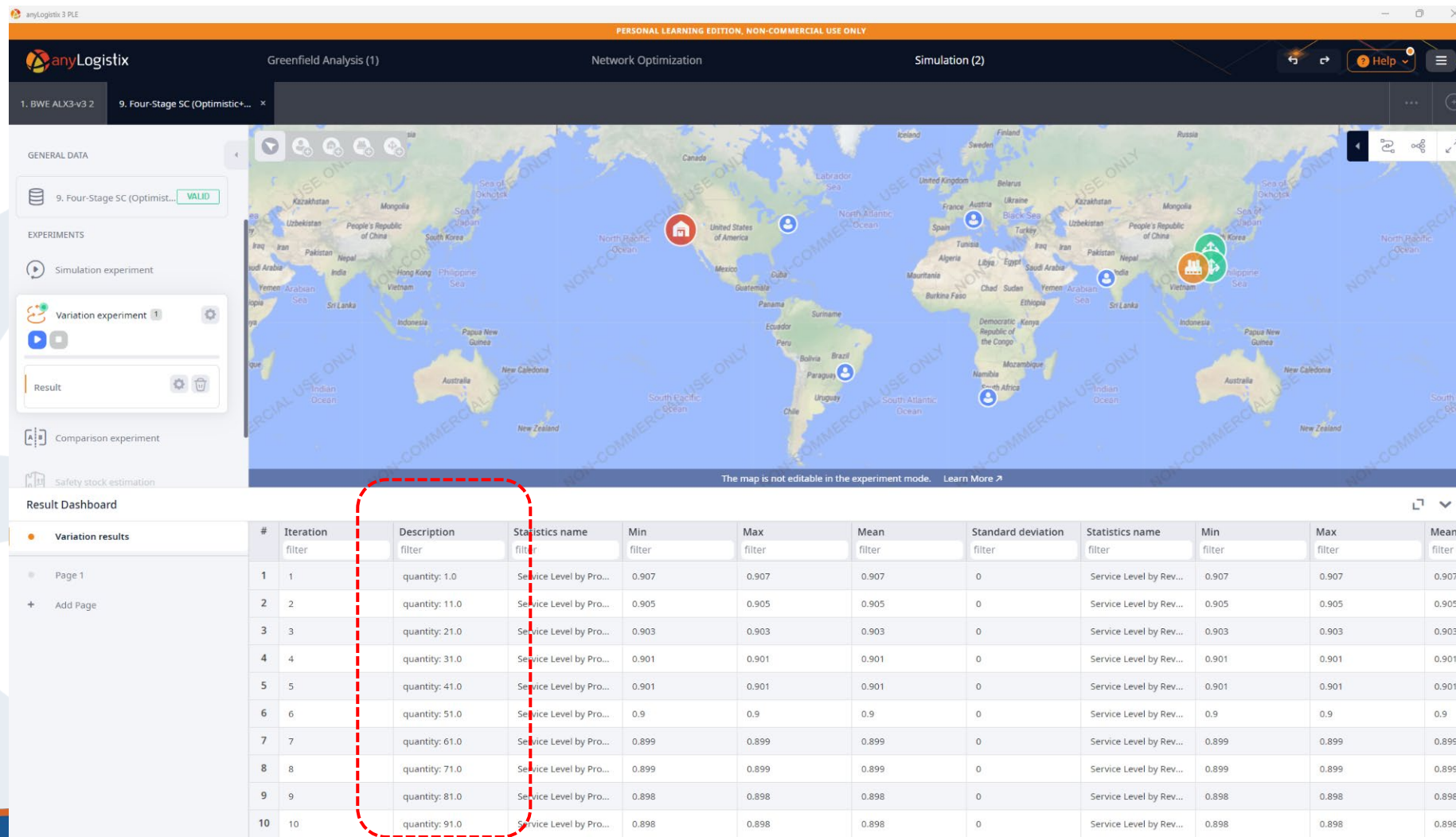


The screenshot displays the anyLogistix 3 PLE software interface. The main window shows a 'Greenfield Analysis (1)' project with a 'Network Optimization' goal. A 'Variation experiment 1' is selected, and its settings are shown in a dialog box. The 'Experiment settings' dialog box includes fields for 'Experiment start' and 'Experiment end', both set to 'All periods'. It also shows 'Use replications' as a toggle switch, 'Replications per iteration' set to 10, and 'Number of threads to use' set to 3. Under 'Variable parameters', there is a list with 'Demand: Demand for customer: US, ...'. The 'Finances stats unit' is set to 'USD', 'Product stats unit' to 'm³', 'Time stats unit' to 'day', and 'Distance stats unit' to 'km'. On the right side of the interface, a list of KPIs is displayed, each with a toggle switch to enable or disable it. The KPIs include: Revenue by Customer, Service Level by Orders, Service Level by Products, Service Level by Revenue, Shipment Schedule Table, Shipments Table, Shipped Vehicles, Site Closure Cost, Staff Busy Time (DC with Staff), Staff Idle Time (DC with Staff), Staff Utilization (DC with Staff), Tariffs, Total CO2, Total Cost, Total Interest Paid, Transportation Cost, Traveled Distance, Utilized Volume (Yogurt Factory), Vehicle Capacity Utilization, Vehicle Loading Time, and Vehicle Unloading Time.



Demand variation scenario

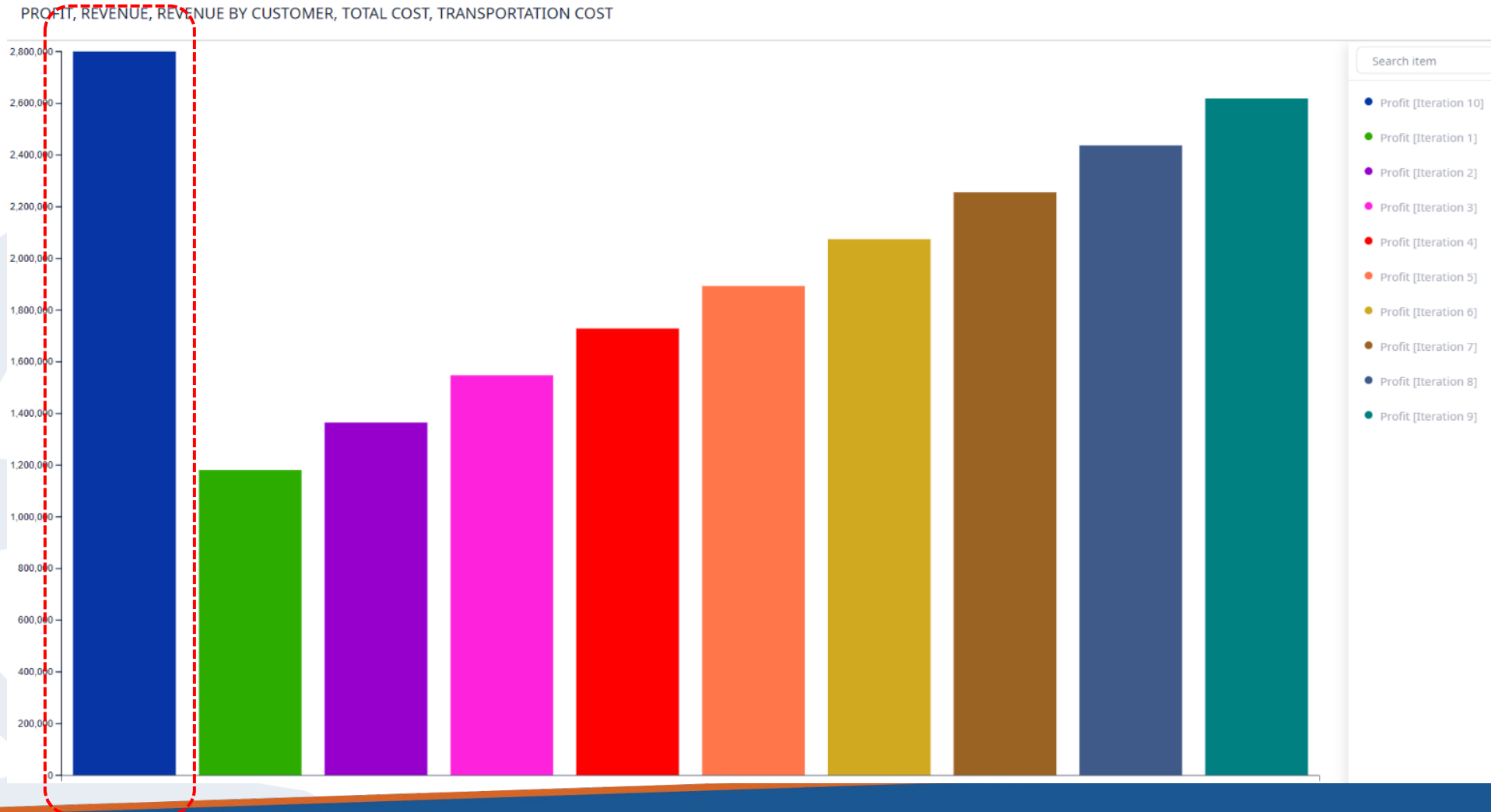
- We have generated 10 experiments in which the US demand has gone through the values 1,11,21,31,41...91.





Demand variation scenario

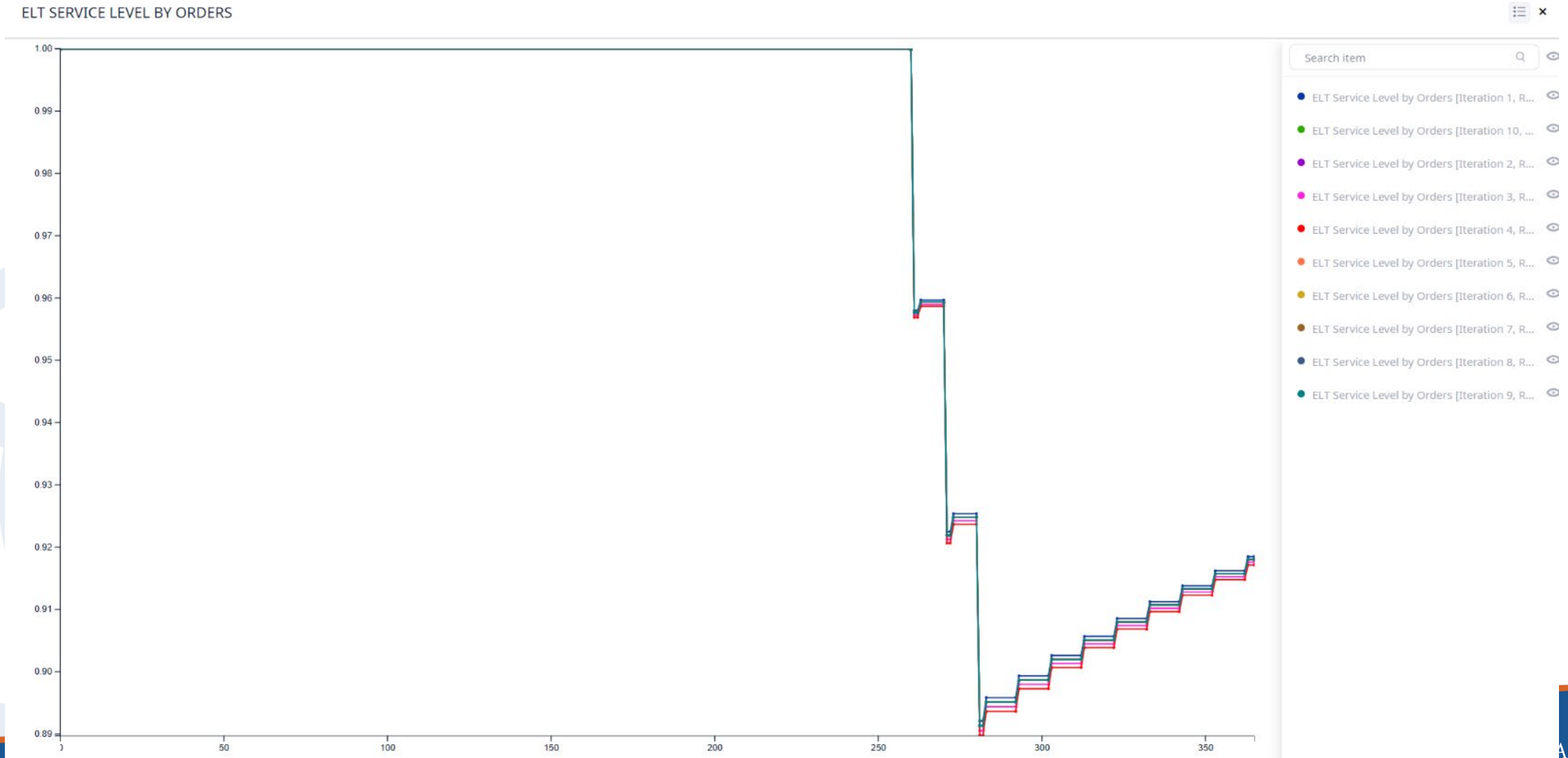
- What is the most profitable scenario? → KPI=profit





Demand variation scenario

- What is the best service scenario? → KPI = ELT service level by order





Exercise 11: Transport costs variation.

- Transport costs bounding
- Variation experiments



Variations in international air transportation

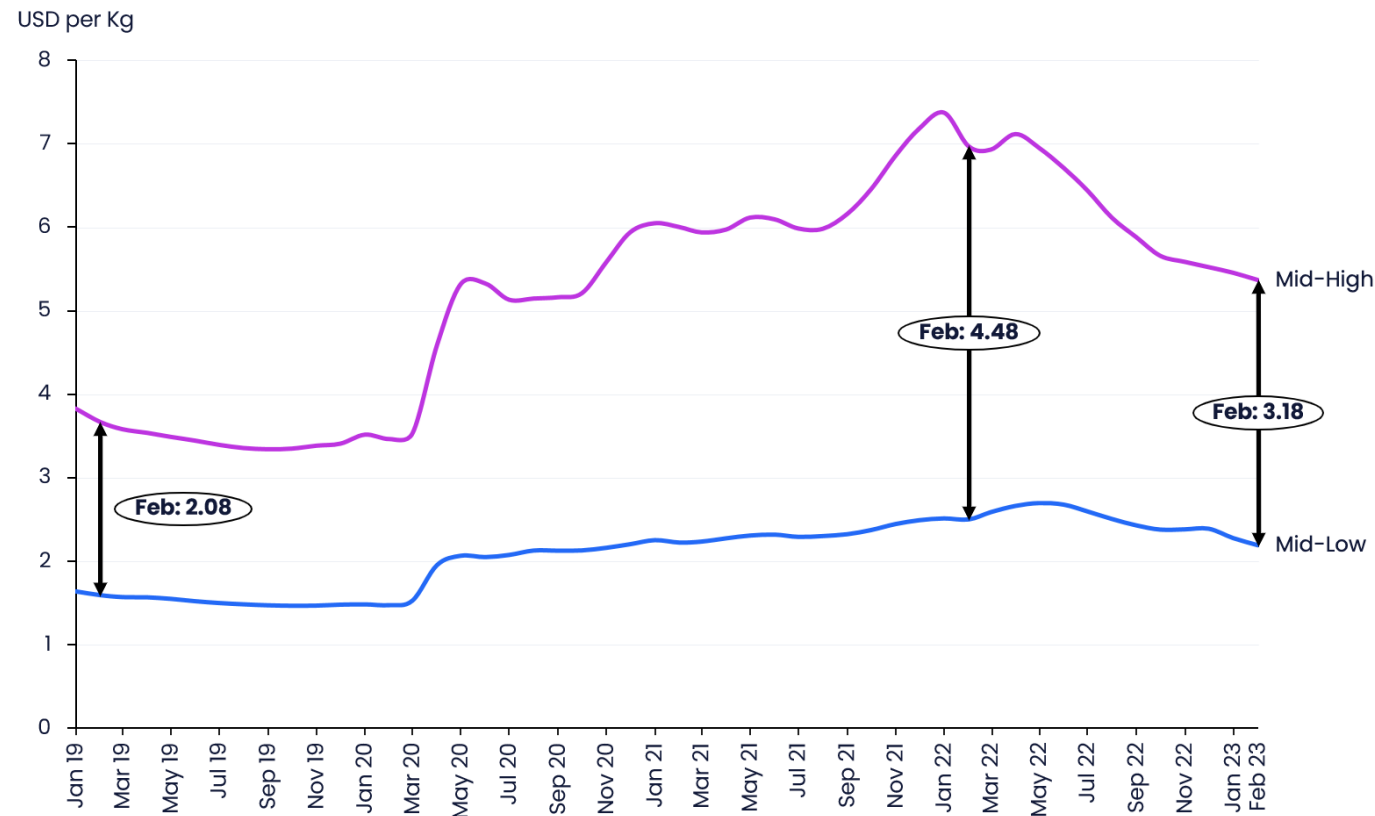


Can you tell me what impact this will have on our income statement?

- The strategy department passes this report on to us:

The spread of airline global air freight mid-low and mid-high rates

(Mid-low: the rate at which 25% of market rates falls below; Mid-high: the rate at which 75% of market rates falls below)





Transportation cost variation

- We recall the current cost scheme:

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anyLogistix

Greenfield Analysis (1) Network Optimization Simulation (2)

1. BWE ALX3-v3 2 9. Four-Stage SC (Optimistic+... x

GENERAL DATA

9. Four-Stage SC (Optimistic+... CHECK

EXPERIMENTS

- Simulation experiment
- Variation experiment 1
- Comparison experiment
- Safety stock estimation
- Risk analysis experiment

Basic All In Use (16)

filter

#	From	To	Cost Calculation	Cost Calculation Parameters	CO2 C	Currency	Distance	Transportation Time	Distance Unit	Time Unit	Straight	Vehicle Type
	filter	filter	filter			filter	filter		filter	filter	filter	filter
1	Supplier China locat	Factory location	Distance-based with	$0.5 * \text{distance} + 0 \text{ per stop}$	0 *...	USD	0	0	km	day	☐	Truck
2	Supplier Taiwan loci	Factory location	Distance-based with	$0.8 * \text{distance} + 0 \text{ per stop}$	0 *...	USD	0	0	km	day	☐	Ferry
3	Factory location	DC location	Product&distance-b	$0.01 * \text{product (m}^3) * \text{distance}$	0 *...	USD	0	2	km	day	☒	Airplane
4	DC location	(All locations)	Product&distance-b	$0.01 * \text{product (m}^3) * \text{distance}$	0 *...	USD	0	2	km	day	☒	Airplane

Locations (9)

Paths (4)

Periods (1)

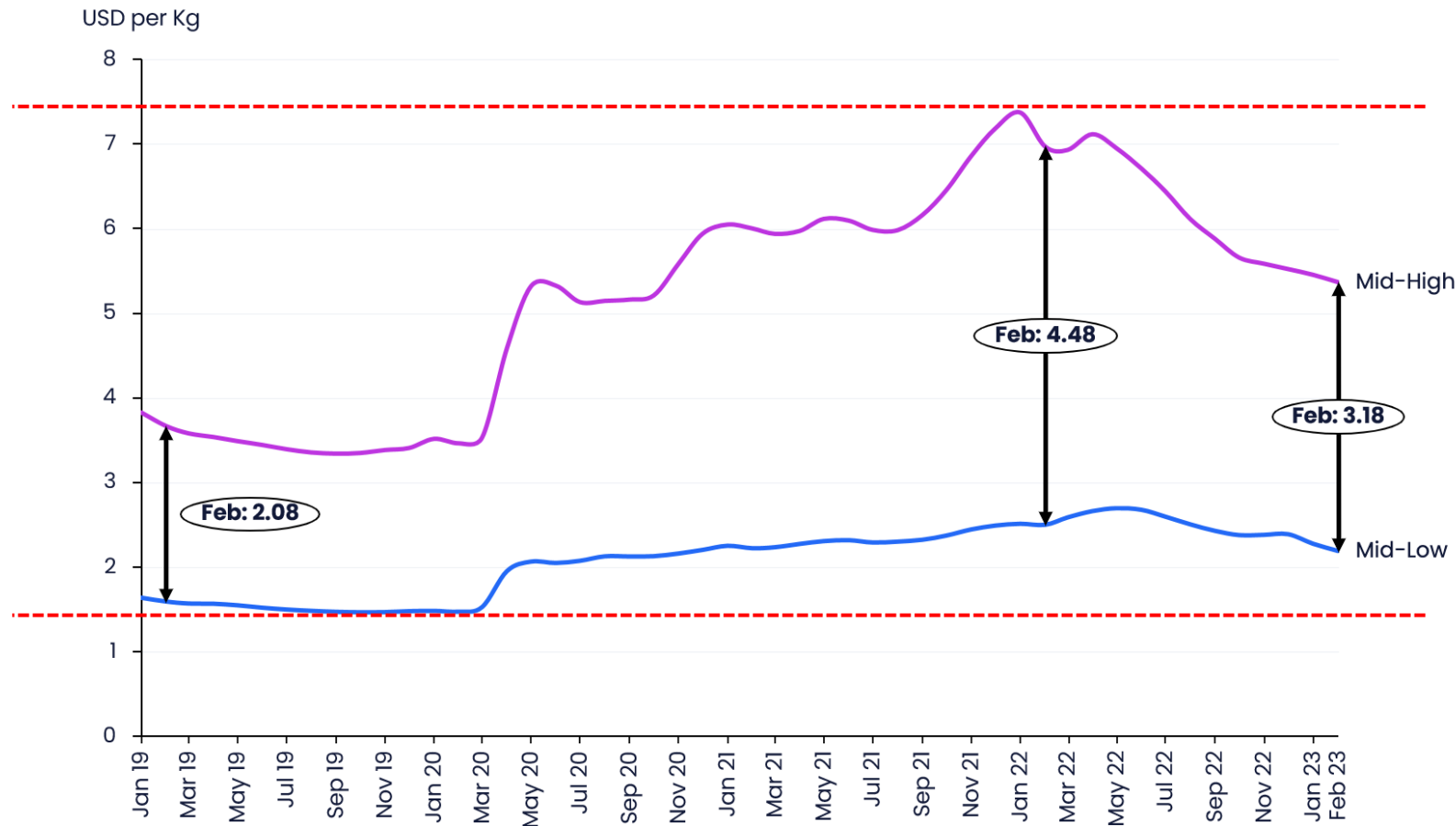


Select the variation band

- We get 2 bands (25th and 75th percentiles)
 - 75% between 3.5 and 7.1
 - 25% between 1.5 and 2.2.
- If we want to have a good sensitivity analysis, we take the minimum and the maximum:
 - Between 1.5 and 7.1

The spread of airline global air freight mid-low and mid-high rates

(Mid-low: the rate at which 25% of market rates falls below; Mid-high: the rate at which 75% of market rates falls below)



Source: Xeneta



Transportation cost variation

- We have to vary all routes that use the aircraft.
- And we have to adapt the scale of our experiment:
 - We scale by our factor (0.01) $\rightarrow$ 0.015 and 0,071
 - We divide in equal periods, for example if we want to have 10 experiments we can step at 0.0056 $(0,071-0,015)/10$

The screenshot shows the anyLogistix software interface. The main window displays a list of experiments under the 'EXPERIMENTS' section, including 'Simulation experiment', 'Variation experiment' (highlighted with a red dashed arrow), 'Comparison experiment', 'Safety stock estimation', and 'Risk analysis experiment'. The 'Variation experiment' is selected, and its settings are shown in the 'Experiment settings' dialog box. The dialog box includes fields for 'Experiment start' and 'Experiment end', both set to 'All periods'. It also has a 'Use replications' toggle switch, 'Replications per iteration' set to 10, and 'Number of threads to use' set to 3. Under 'Variable parameters', there are two entries: 'Paths: Path: [Factory location - DC loc...]' and 'Paths: Path: [DC location - (All locatio...]'.



Execute the variation experiment...

- We obtain the expected 10×10 variations.

anyLogistix 3 PLE

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anyLogistix Greenfield Analysis (1) Network Optimization Simulation (2)

1. BWE ALX3-v3 2 9. Four-Stage SC (Optimistic+...

Result Dashboard

Variation results

Page 1

+ Add Page

#	Iteration	Description	Statistics name	Min	Max	Mean	Standard deviation	Statistics name	Min	Max
70	70	costPerUnit: 0.043, costPerUnit: 0.054	filter Service Level by Ord...	filter 0.897	filter 0.897	filter 0.897	filter 0	filter Service Level by Pro...	filter 0.901	filter 0.901
77	77	costPerUnit: 0.049, costPerUnit: 0.054	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
78	78	costPerUnit: 0.054, costPerUnit: 0.054	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
79	79	costPerUnit: 0.06, costPerUnit: 0.054	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
80	80	costPerUnit: 0.065, costPerUnit: 0.054	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
81	81	costPerUnit: 0.015, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
82	82	costPerUnit: 0.021, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
83	83	costPerUnit: 0.026, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
84	84	costPerUnit: 0.032, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
85	85	costPerUnit: 0.037, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
86	86	costPerUnit: 0.043, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
87	87	costPerUnit: 0.049, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
88	88	costPerUnit: 0.054, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
89	89	costPerUnit: 0.06, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
90	90	costPerUnit: 0.065, costPerUnit: 0.06	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
91	91	costPerUnit: 0.015, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
92	92	costPerUnit: 0.021, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
93	93	costPerUnit: 0.026, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
94	94	costPerUnit: 0.032, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
95	95	costPerUnit: 0.037, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
96	96	costPerUnit: 0.043, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
97	97	costPerUnit: 0.049, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
98	98	costPerUnit: 0.054, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
99	99	costPerUnit: 0.06, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901
100	100	costPerUnit: 0.065, costPerUnit: 0.065	Service Level by Ord...	0.897	0.897	0.897	0	Service Level by Pro...	0.901	0.901



Variations in international air transportation



- Our transportation costs will range from

TRANSPORTATION COST			
#	Statistics	Iteration	Value
	filter	filter	filter
1	Transportation Cost	Iteration 1	1,054.374

TRANSPORTATION COST			
#	Statistics	Iteration	Value
	filter	filter	filter
1	Transportation Cost	Iteration 100	4,597.071

- And the benefit, between:

TRANSPORTATION COST			
#	Statistics	Iteration	Value
	profit ×	filter	filter
1	Profit	Iteration 100	1,780,652.929

TRANSPORTATION COST			
#	Statistics	Iteration	Value
	profit ×	filter	filter
1	Profit	Iteration 1	1,784,195.626

We have almost no risk: 0.19% impact on profit.

A brief statistics recap

- Risk requires some statistical attention.



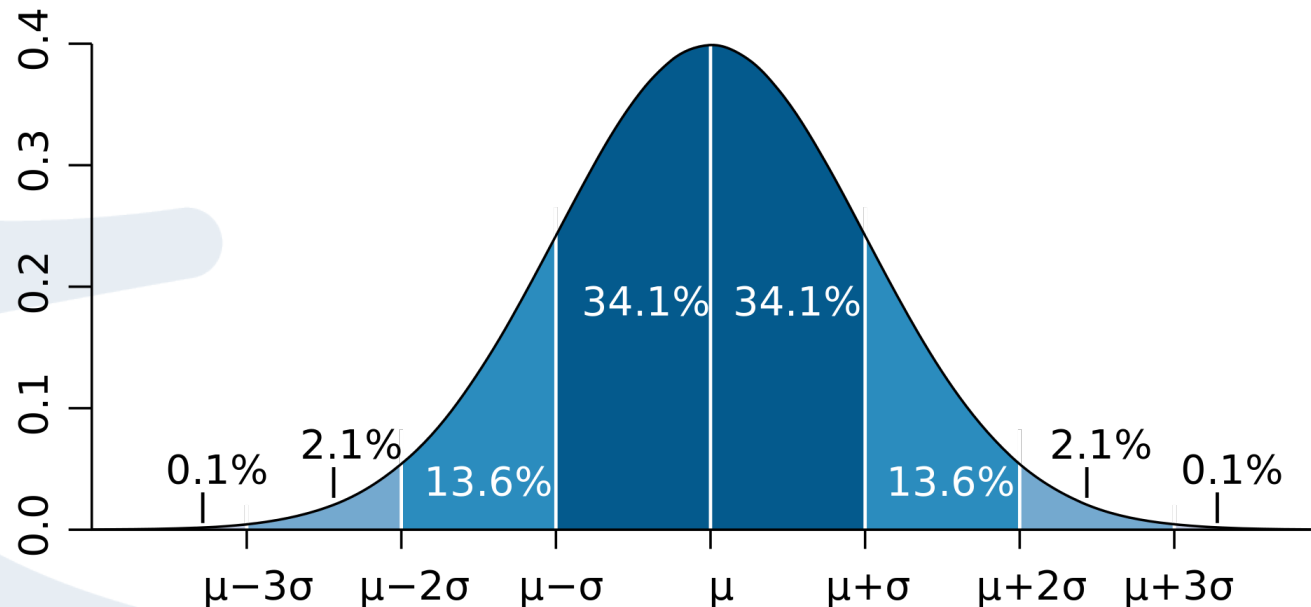
Risk = probability

- Risk is represented by probability distributions.
- Probability of what?
 - Probability of occurrence → events
- How to distinguish whether an experimental result is genuine, structural, or simply the result of chance?
 - Example: Demand is normally an estimate from the past into the future, but is usually described by a probability function.
 - Normal function, Gaussian, with mean μ and standard deviation



Random, it can be random, but it can also be structural.

- "Madrid's demand for a certain product is 100 units per month on average, and a standard deviation of 10."
 - Clearly, there is uncertainty as to the specific values.
 - But it is also clear that **certain ranges of values are very likely to occur.**
 - But it is also clear that extreme values can occur, albeit with very low probability.





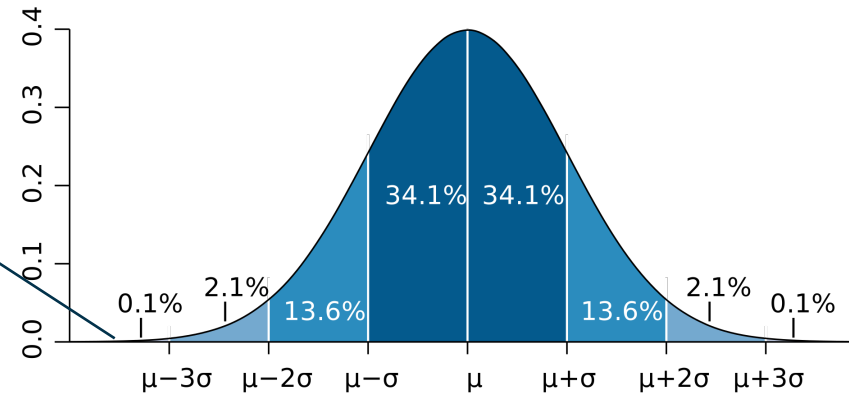
Statistically significant experiment

- Relevant experiment = "statistically significant".
 - One in which the results are sufficiently different from what would be expected by chance, suggesting that there is a real relationship between the variables studied or a genuine effect of the intervention or treatment applied.
- The significance level indicates how improbable the result must be under the null hypothesis to be considered significant.
- Confidence level of the results:
 - Commonly, a significance level of 5% (0.05) is used, which means that there is less than a 5% probability that the observed results are the result of chance.
 - This is the same as saying that the experiment has a confidence level of 95%.
- A statistically significant experiment allows conclusions to be drawn and decisions to be made based on its results.

How do you know if an experiment is meaningful?

- Could we draw conclusions from running a single experiment?

Let's imagine that when running the experiment randomly, it took a value that was highly improbable...



Conclusion: the first experiment was NOT significant

... however, if we repeated the experiments more times we would see that the results would be different since 68.2% would be in the dark part, and 95% would be in the following section

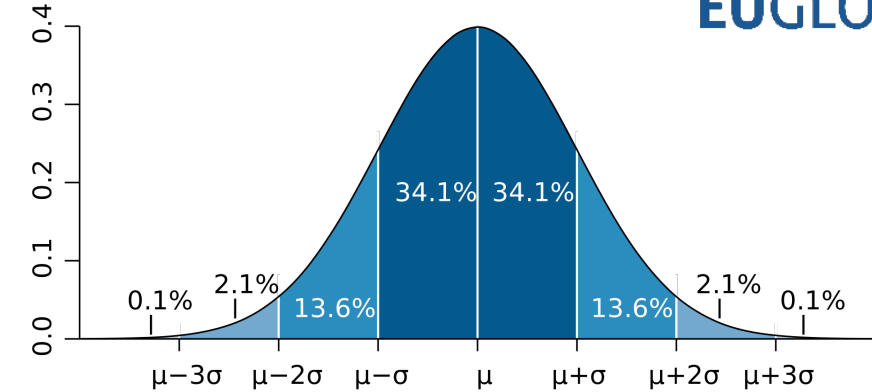


Statistical hypotheses

- NULL hypothesis (H_0):
 - It represents the idea that there is no effect or difference between the conditions or groups being compared. In other words, it states that any observed outcome is simply the result of chance or random variability.
- ALTERNATIVE hypothesis (H_1):
 - It represents the idea that there is some kind of effect or difference between the conditions or groups studied.
 - Inverse to the null hypothesis.
- The objective is to determine statistically whether the results of the simulations are sufficiently different from what would be expected under the null hypothesis (everything is random and nothing structural) to be considered statistically significant.



How do you know if an experiment is meaningful?



- In order to draw conclusions, it is necessary to **repeat** the probabilistic risk estimation experiments many times.
- If we collect the results of the repetitions and extract their mean μ and standard deviation we can evaluate their confidence level.
- Not to be confused with variation experiments:
 - In VARIATION experiments, the parameters are modified according to a range.
 - In REPEAT experiments the same experiment is performed many times taking into account the random values of some variables.
- In Anylogistix the repetitions are called "Replications".



Exercise 12: risks and experiment iteration.



Back to WHC-Mobile



Let's continue to analyze WHC's supply chain disruption from a RISK ANALYSIS point of view.



Supplier Taiwan, Supplier China



Factory



DC



(All customers)





Initial situation

- Let's use the initial scenario of supply chain breakage with absolute probability "1".

anyLogistix 3 PLE

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anyLogistix Greenfield Analysis (1) Network Optimization Simulation (2)

10. Four-Stage SC (Optimistic-... x Gazelle Factory Risk Analysis

GENERAL DATA

10. Four-Stage SC (Optimistic-... CHECK

EXPERIMENTS

- Simulation experiment
- Variation experiment
- Comparison experiment

Basic All In Use (16)

filter

#	Name	Event Type	Parameters	Occurrence Type	Occurrence Time	Trigger	Probability
1	DC-USA: Fire	Facility state	Object: DC-USA, New ...	Date	8/10/17 12:00 AM		1
2	DC-USA: Full recovery	Facility state	Object: DC-USA, New ...	Delay (days)	30	DC-USA: Fire	1

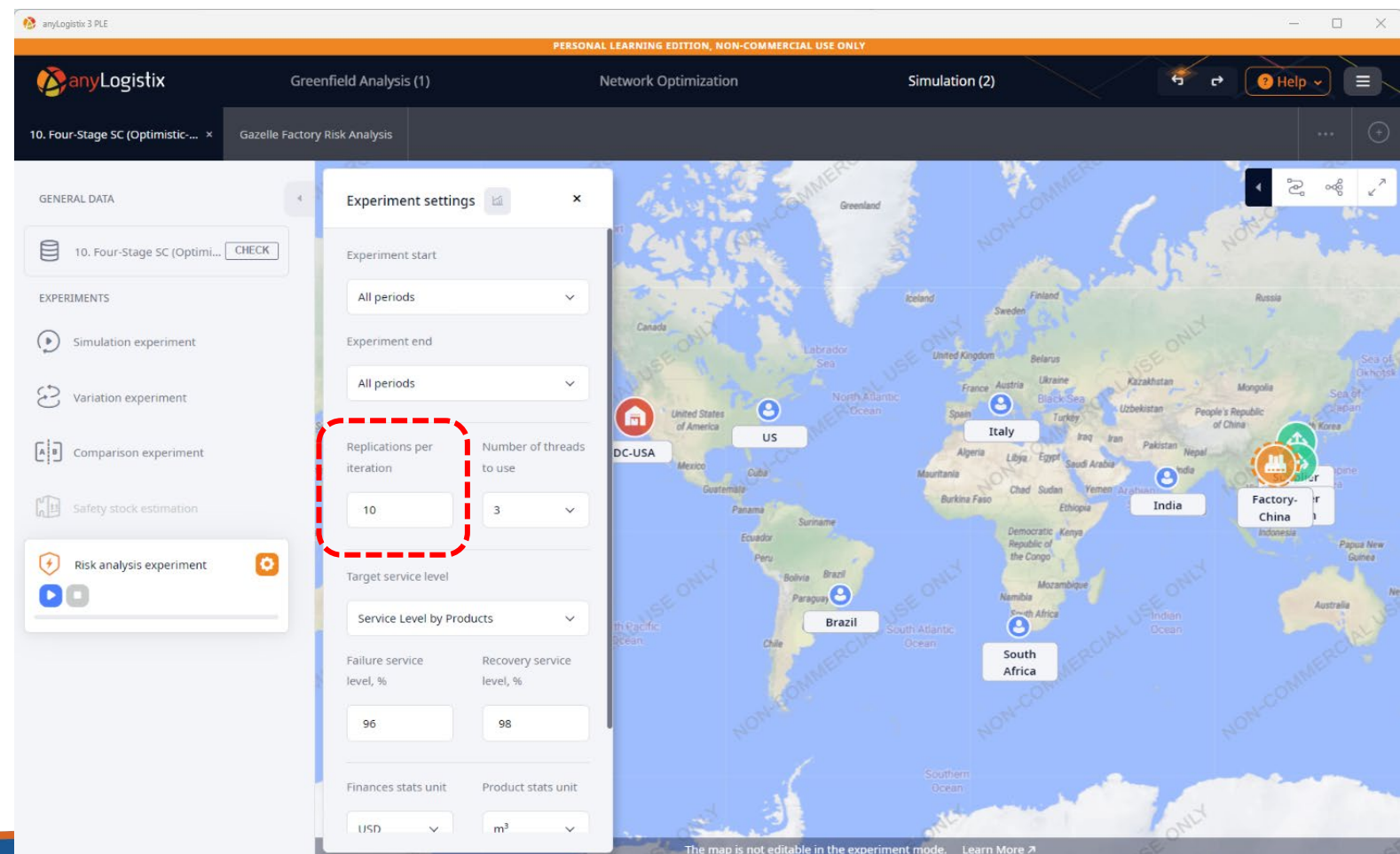
Events (2)

- Facility Expenses
- Fleets



Basic risk analysis

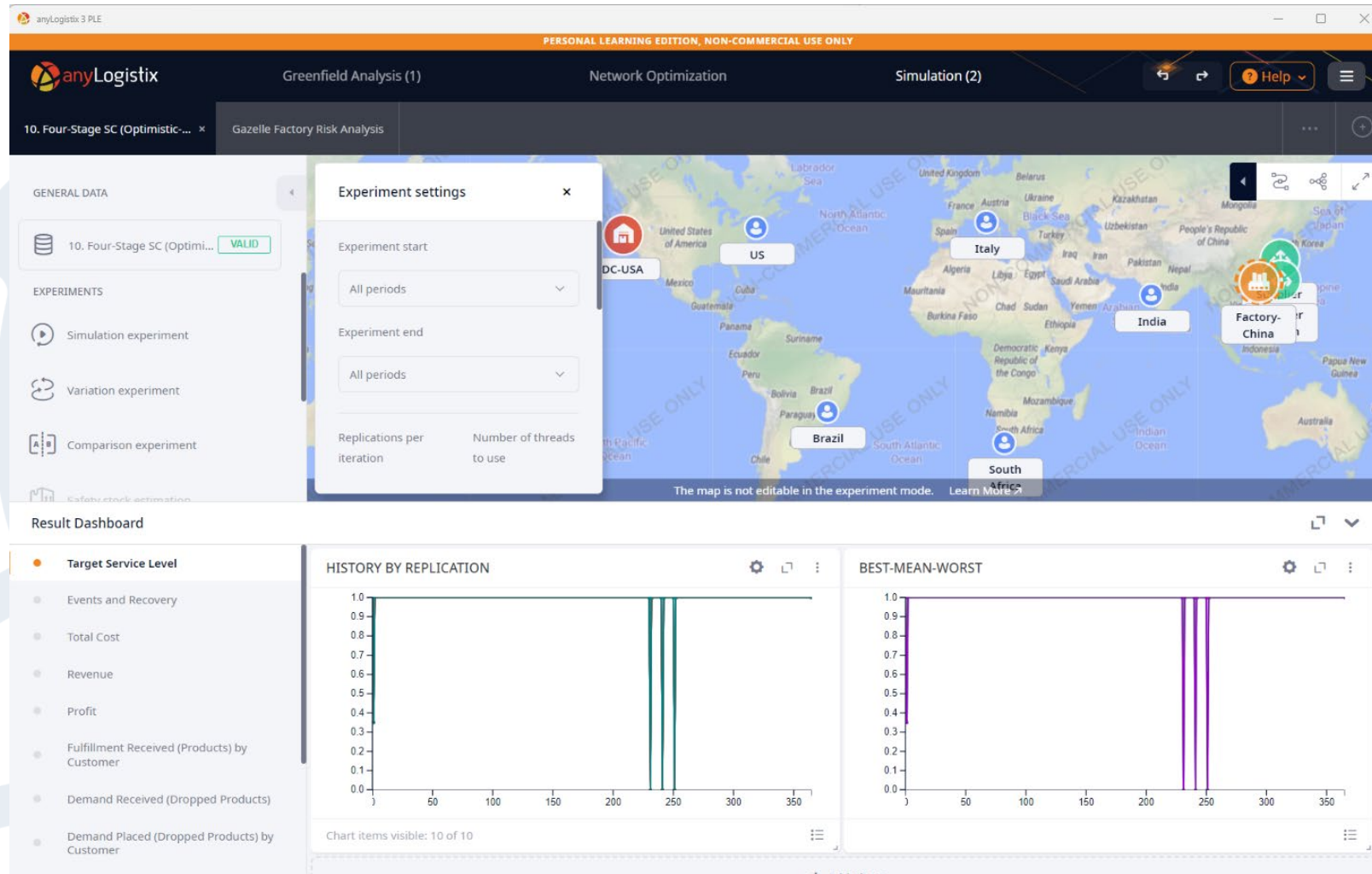
- We can ask ALX to run the SC break experiment many times to compare the results.
- Spoiler: all results will be the same, since the experiment is deterministic → the probability is always '1'.



The screenshot displays the anyLogistix 3 PLE software interface. The top navigation bar includes tabs for 'Greenfield Analysis (1)', 'Network Optimization', and 'Simulation (2)'. The main workspace is titled '10. Four-Stage SC (Optimistic...)' and 'Gazelle Factory Risk Analysis'. A 'GENERAL DATA' sidebar on the left lists various experiment types, with 'Risk analysis experiment' selected. The 'Experiment settings' dialog box is open, showing fields for 'Experiment start' and 'Experiment end' (both set to 'All periods'), 'Replications per iteration' (set to 10, highlighted with a red dashed box), 'Number of threads to use' (set to 3), 'Target service level' (set to 'Service Level by Products'), 'Failure service level, %' (set to 96), 'Recovery service level, %' (set to 98), 'Finances stats unit' (set to USD), and 'Product stats unit' (set to m³). The background features a world map with location markers for DC-USA, US, Italy, India, Brazil, and South Africa.



Results



- Depending on the variables (KPIs) that we have selected to be included in the simulation, we obtain the corresponding scorecards.



Results, determinism

anyLogistic 3 PLE

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anyL

EVENTS TABLE

#	Replication	Event	Date	Day #
	filter	filter	filter	filter
1	Replication 1	DC-USA: Fire	2017-08-10T00:00	221
2	Replication 1	DC-USA: Full recovery	2017-09-09T00:00	251
3	Replication 10	DC-USA: Fire	2017-08-10T00:00	221
4	Replication 10	DC-USA: Full recovery	2017-09-09T00:00	251
5	Replication 2	DC-USA: Fire	2017-08-10T00:00	221
6	Replication 2	DC-USA: Full recovery	2017-09-09T00:00	251
7	Replication 3	DC-USA: Fire	2017-08-10T00:00	221
8	Replication 3	DC-USA: Full recovery	2017-09-09T00:00	251
9	Replication 4	DC-USA: Fire	2017-08-10T00:00	221
10	Replication 4	DC-USA: Full recovery	2017-09-09T00:00	251
11	Replication 5	DC-USA: Fire	2017-08-10T00:00	221
12	Replication 5	DC-USA: Full recovery	2017-09-09T00:00	251
13	Replication 6	DC-USA: Fire	2017-08-10T00:00	221
14	Replication 6	DC-USA: Full recovery	2017-09-09T00:00	251
15	Replication 7	DC-USA: Fire	2017-08-10T00:00	221
16	Replication 7	DC-USA: Full recovery	2017-09-09T00:00	251

Close

10. Four-Stage

GENERAL DATA

10. Four-Stage

EXPERIMENTS

Simula

Variati

Compe

Result Dash

Target

Events

Total C

Reven

Profit

Fulfilme

Custom

Demand Received (Dropped Products)

5 Replication 1 DC-USA: Fire 2017-08-10T00:00 221

6 Replication 1 DC-USA: Full recovery 2017-09-09T00:00 251

5 Replication 3 Smartphone 2017-08-10T00:00 221

6 Replication 3 Smartphone 2017-09-09T00:00 251

- If we open the events table, we see that in all iterations the Fire and Recovery events are always the same.

→ All experiments are the same.



Results: history, best-m

In this case, all are equal, because the experiment is deterministic.

Result Dashboard

- Target Service Level
- Events and Recovery
- Total Cost
- Revenue
- Profit**
- Fulfillment Received (Products) by Customer
- Demand Received (Dropped Products)
- Demand Placed (Dropped Products) by Customer

HISTORY BY REPLICATION

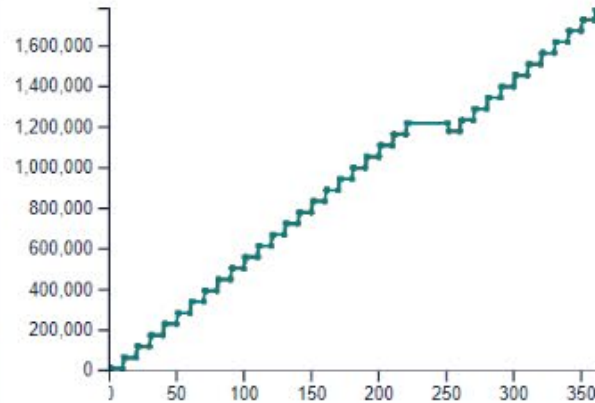
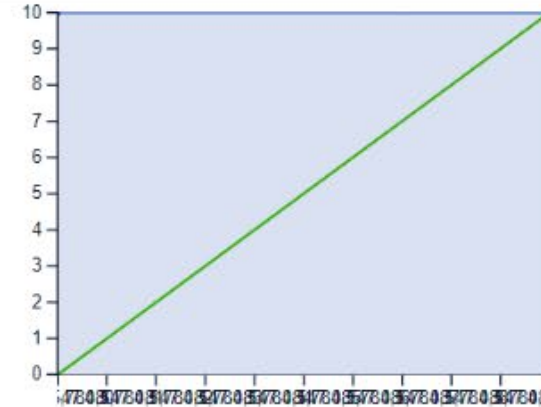
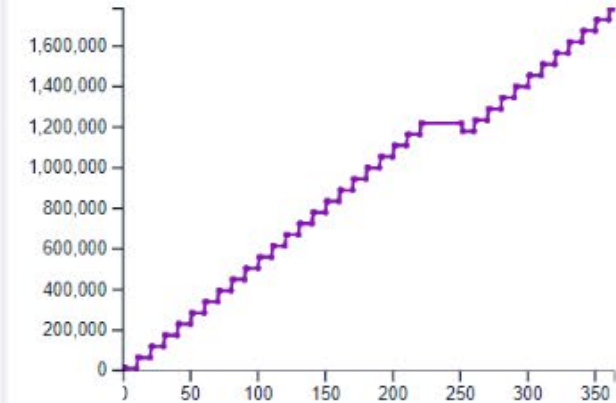


Chart items visible: 10 of 10

HISTOGRAM



BEST-MEAN-WORST



- All KPIs collected are shown as:
 - Simulation history:** all runs are compared with each other to see how they evolve over time.
 - Histogram:** the distributions of occurrence of the final values are shown, to see the relevance of the experiments → mean and deviation.
 - Best-Average-Worst-Best:** the best, worst and average KPI values are shown.



Exercise 13: Randomness - Uniform.



Ex.11. Probability of occurrence of events



The probability of the Fire occurring is 0.5.
The probability of recovering from the Fire is 0.5



Supplier Taiwan, Supplier China



Factory



DC



(All customers)





Ex.11 Setting the probability

Risk analysis experiment 2

Basic All In Use (16)

filter

DCs and Factories (2)

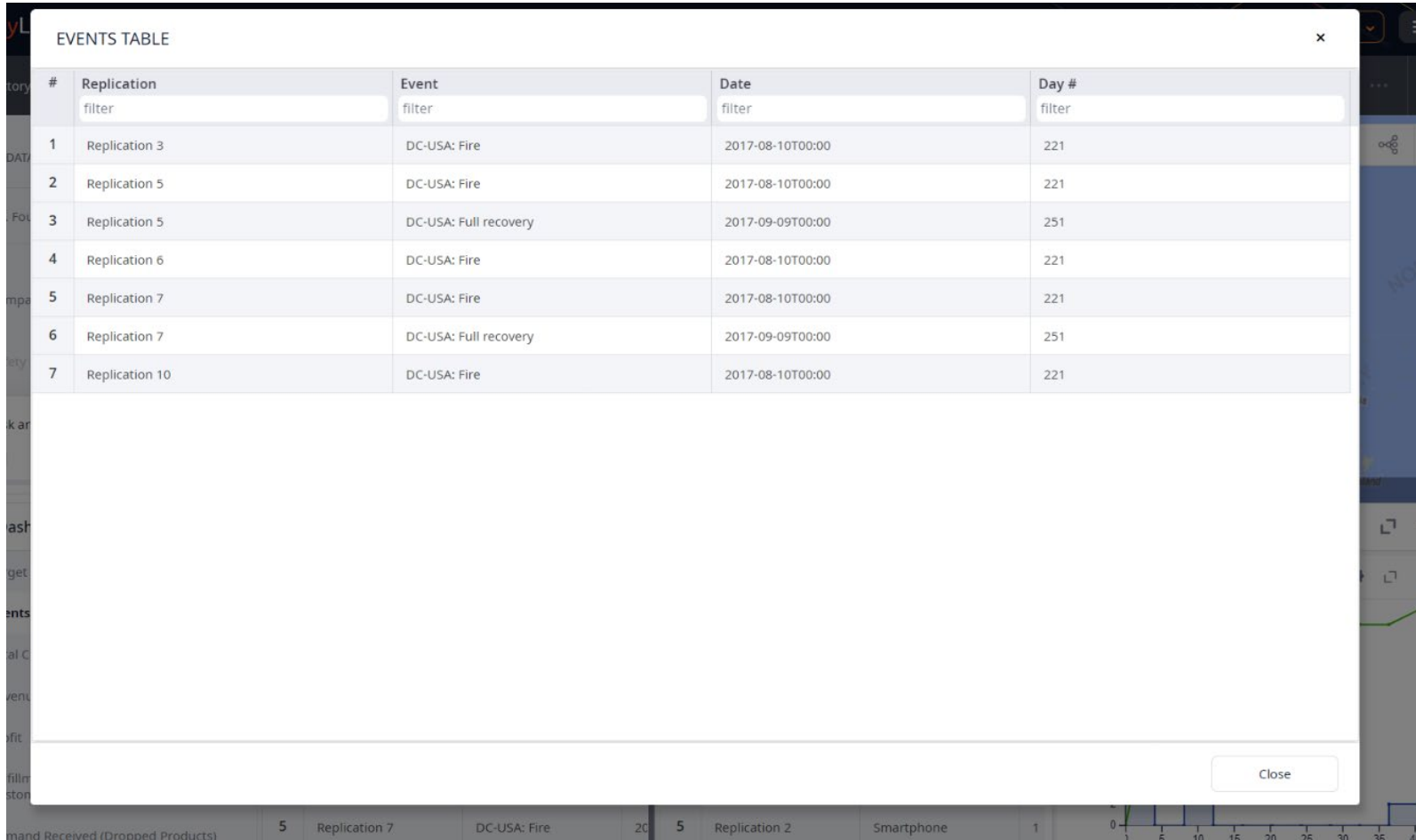
Demand (5)

#	Name	Event Type	Parameters	Occurrence Type	Occurrence Time	Trigger	Probability
1	DC-USA: Fire	Facility state	Object: DC-USA, New ...	Date	8/10/17 12:00 AM		0.5
2	DC-USA: Full recovery	Facility state	Object: DC-USA, New ...	Delay (days)	30	DC-USA: Fire	0.5



Ex.11 - We look at the events

- In this run only iterations 5 and 7 have completed the occurrence of both events.
- In iterations 3, 6 and 10 it did not recover from the fire.
- CAUTION: as everything is random, each time we run the experiment, different values will come out.

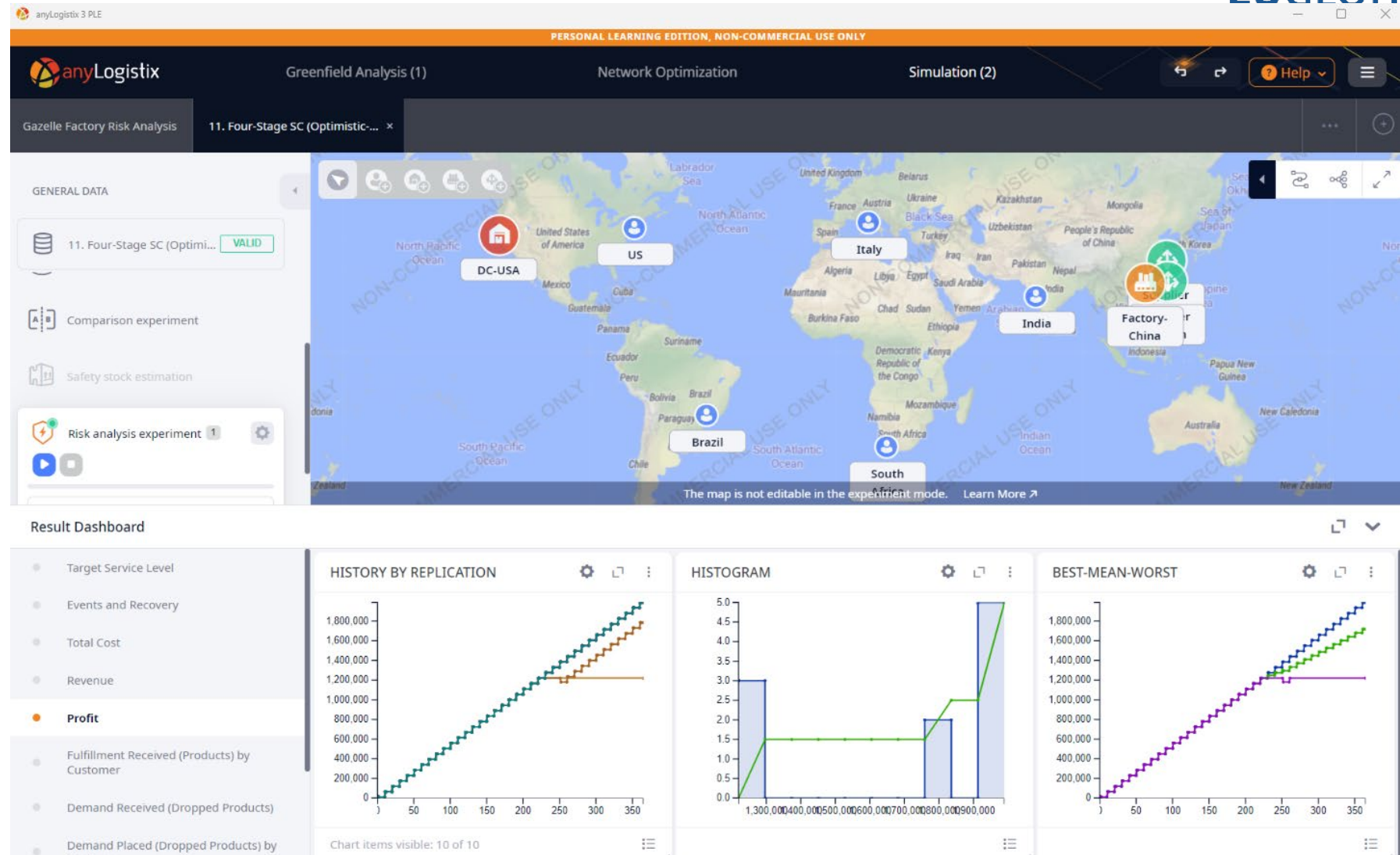


#	Replication	Event	Date	Day #
1	Replication 3	DC-USA: Fire	2017-08-10T00:00	221
2	Replication 5	DC-USA: Fire	2017-08-10T00:00	221
3	Replication 5	DC-USA: Full recovery	2017-09-09T00:00	251
4	Replication 6	DC-USA: Fire	2017-08-10T00:00	221
5	Replication 7	DC-USA: Fire	2017-08-10T00:00	221
6	Replication 7	DC-USA: Full recovery	2017-09-09T00:00	251
7	Replication 10	DC-USA: Fire	2017-08-10T00:00	221



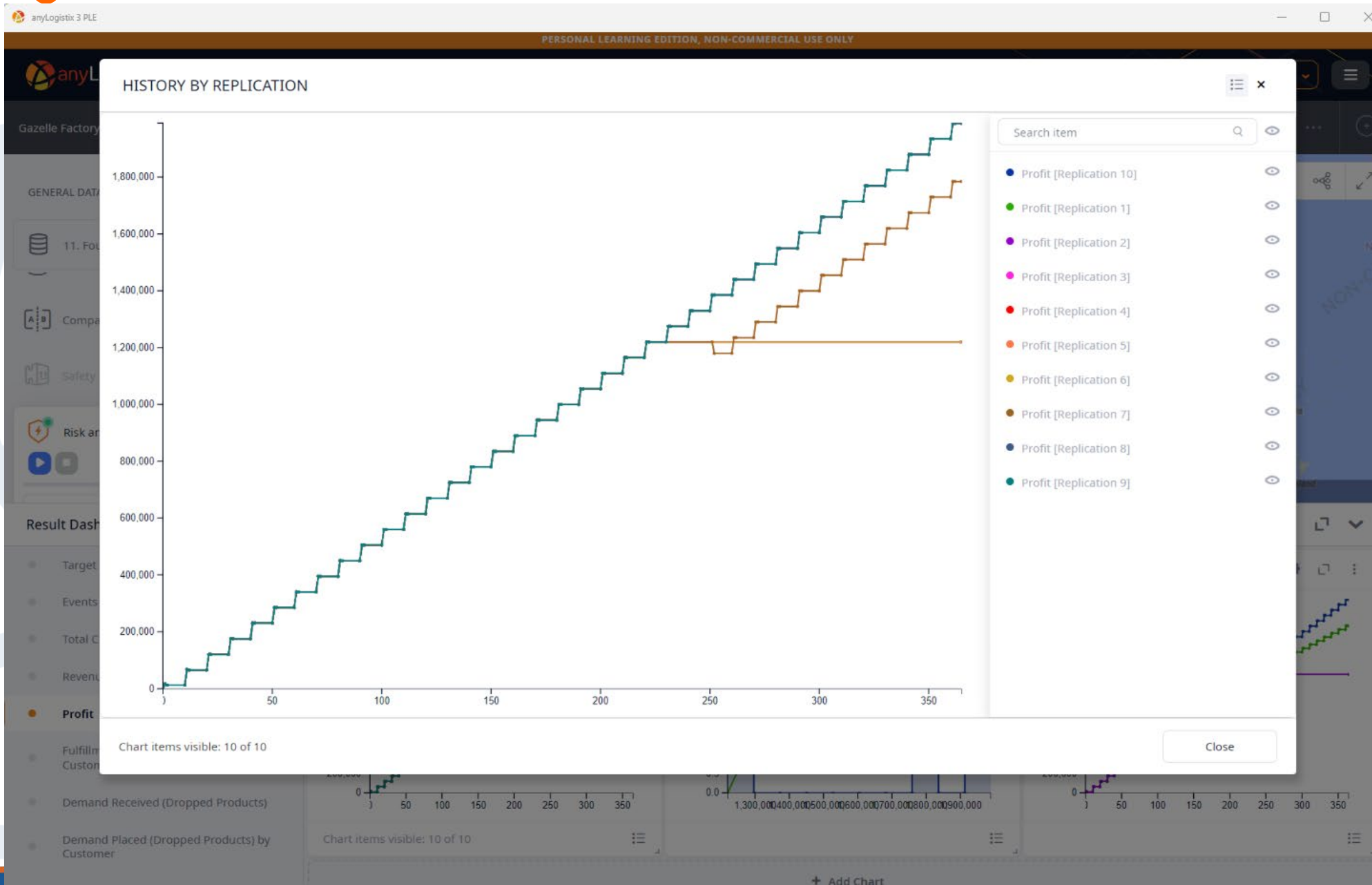
Ex.11 - Analyzing profit

- Here we already see that we have different results.
- The same will occur with the rest of the KPIs under study.



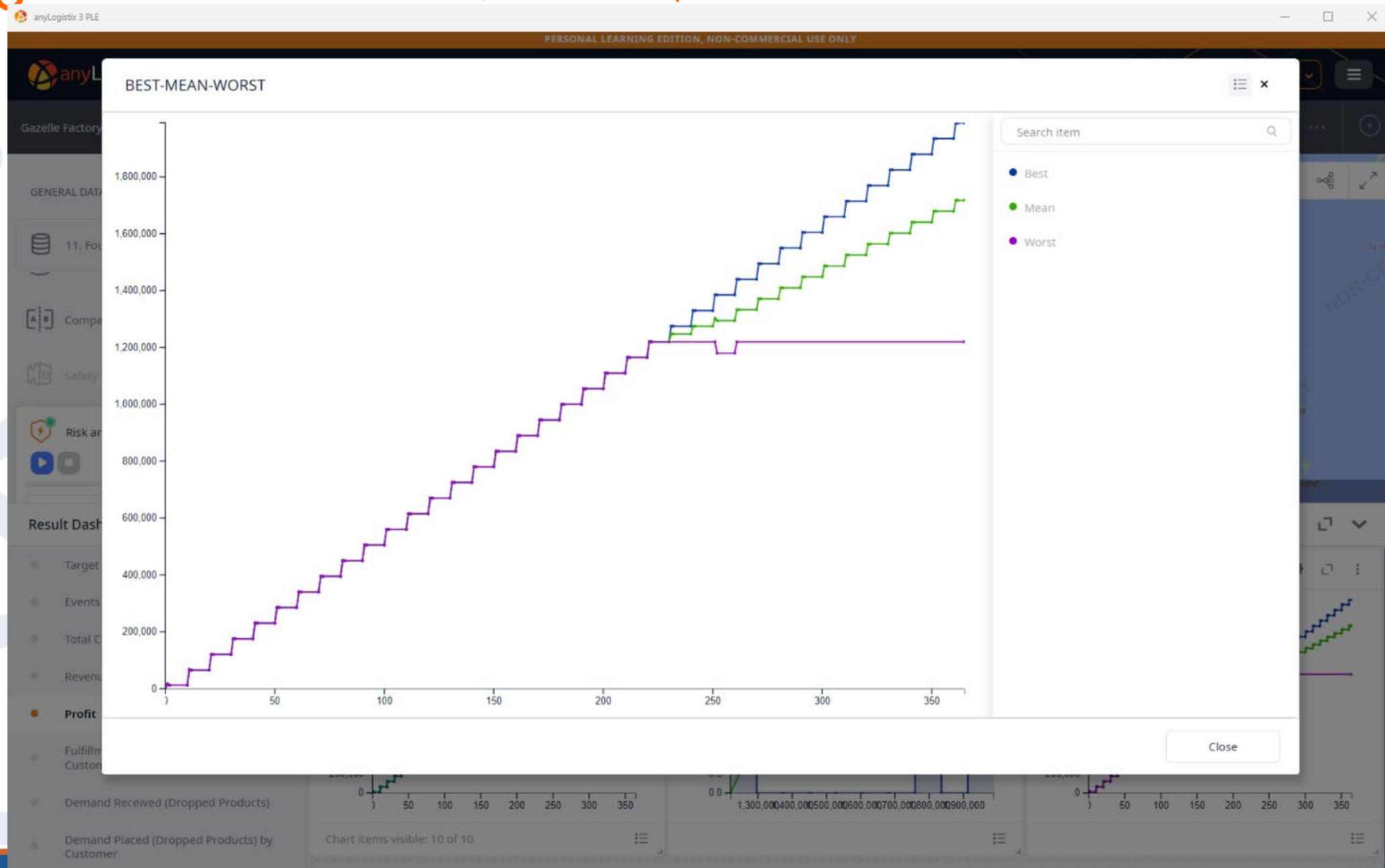


Ex.11 - Analyzing profit





Ex.11 - Analyzing profit





Exercise 14: Randomness - Gaussian.

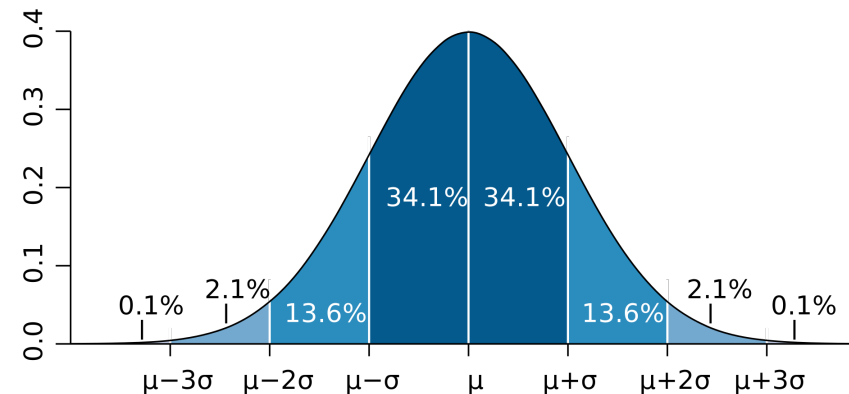


Ex.12 - Events occur with a certain probability distribution



I explained myself badly before... I apologize...

- The probability of fire is RANDOM with probability 0.5.
- However, the duration of recovery follows a normal (Gaussian) distribution of mean 30 and standard deviation 15.





Ex.12 Setting the probabilities

Simulation experiment
Variation experiment
Comparison experiment
Safety stock estimation

Basic **All** In Use (16)

filter

#	Name	Event Type	Parameters	Occurrence Type	Occurrence Time	Trigger	Probability
1	DC-USA: Fire	Facility state	Object: DC-USA, New ...	Random	1/1/17 - 12/31/17		0.5
2	DC-USA: Full recovery	Facility state	Object: DC-USA, New ...	Delay (days)	Normal(15; 30)	DC-USA: Fire	1

- (We could also have configured the probability of recovery to be 0.5 and if so, the time distribution to be normal (30,15). This is not the case...)



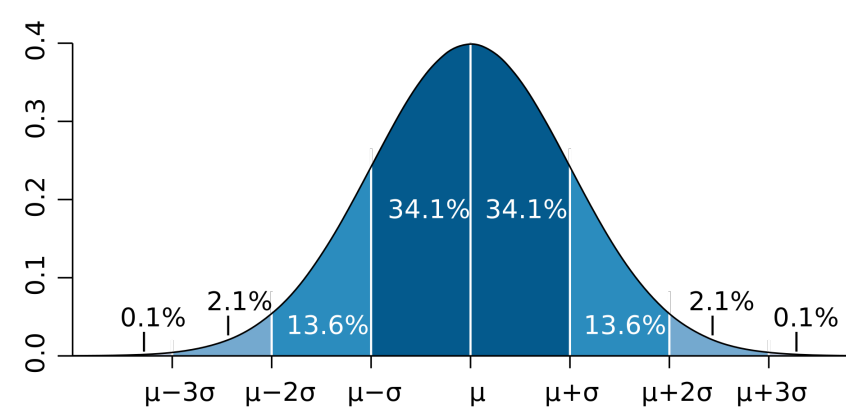
Ex.12 - We look at the events

We interpret the experimental scenarios:

1. Failure to recover from the fire of $t=357$
2. No fire.
3. Fire at $t=101$ and recovery at $t=189$.
4. Fire at $t=100$ and recovery at $t=124$.
5. Fire at $t=176$ and recovery at $t=226$.
6. No fire.
7. No fire.
8. Fire at $t=116$ and recovery at $t=143$.
9. Fire at $t=69$ and recovery at $t=111$.
10. Fire at $t=251$ and recovery at $t=290$.

EVENTS TABLE

#	Replication filter	Event filter	Date filter	Day # filter
1	Replication 1	DC-USA: Fire	2017-12-23T00:31:34.638	357
2	Replication 3	DC-USA: Fire	2017-06-10T10:58:41.165	161
3	Replication 3	DC-USA: Full recovery	2017-07-08T08:57:51.386	189
4	Replication 4	DC-USA: Fire	2017-04-10T18:16:01.271	100
5	Replication 4	DC-USA: Full recovery	2017-05-04T20:47:05.972	124
6	Replication 5	DC-USA: Fire	2017-06-25T09:40:33.335	176
7	Replication 5	DC-USA: Full recovery	2017-08-14T17:08:59.104	226
8	Replication 8	DC-USA: Fire	2017-04-26T15:12:15.437	116
9	Replication 8	DC-USA: Full recovery	2017-05-23T11:23:09.544	143
10	Replication 9	DC-USA: Fire	2017-03-10T21:14:07.892	69
11	Replication 9	DC-USA: Full recovery	2017-04-21T20:55:17.742	111
12	Replication 10	DC-USA: Fire	2017-09-08T04:26:22.991	251
13	Replication 10	DC-USA: Full recovery	2017-10-17T17:25:42.360	290



I'm starting to get the hang of this randomness thing.

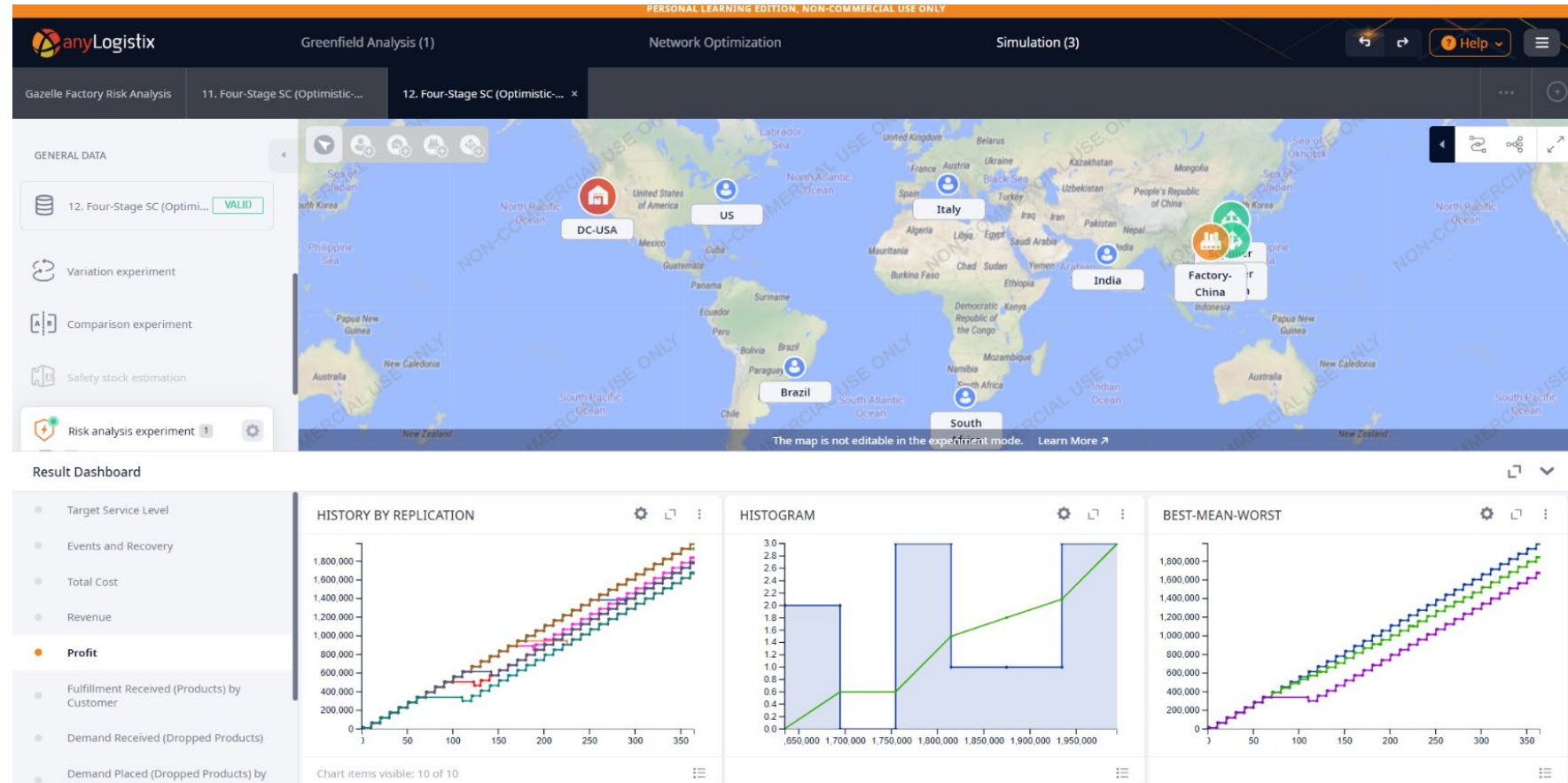
But I am MUCH MORE CONCERNED because, as you are telling me, depending on what happens with the damn Gaussian, my financial situation is in jeopardy!!!!!!

Can you give me more visibility on this?



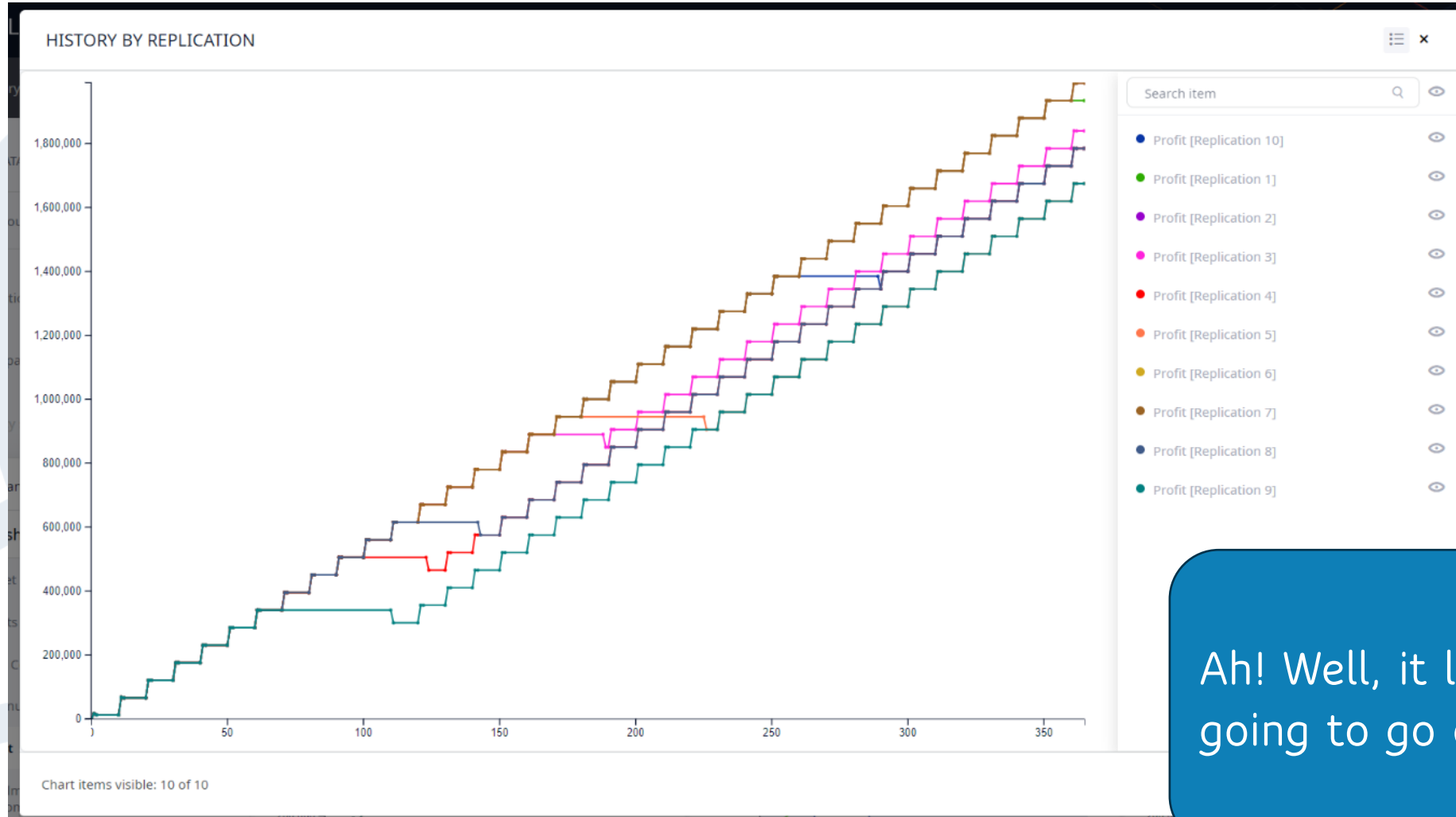
Analyzing financial risk from the point of view of profit

- Now we can study the variability of the profit according to the different experiments performed.





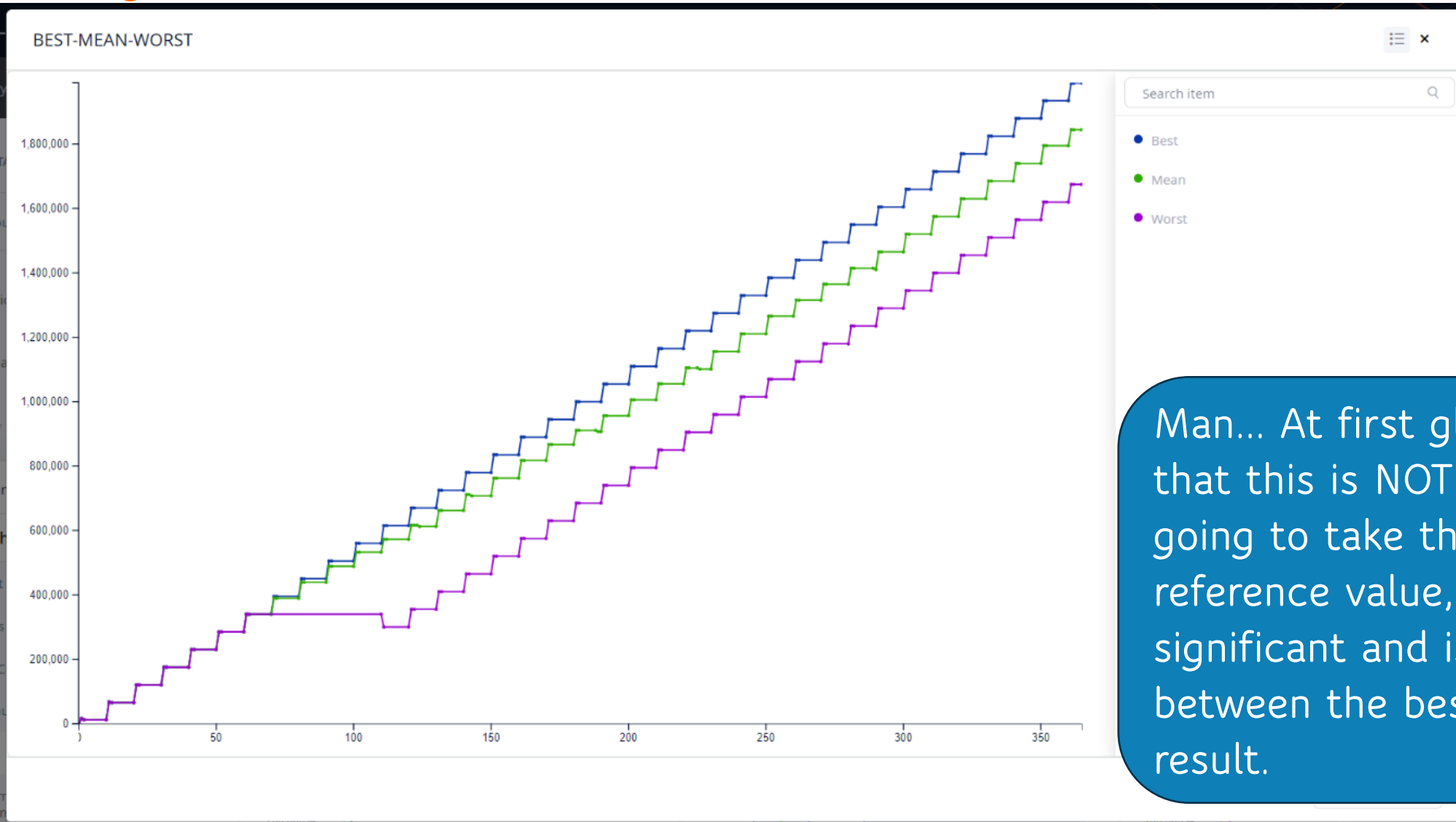
Analyzing profit



Ah! Well, it looks like we're not going to go out of business....



Analyzing profit

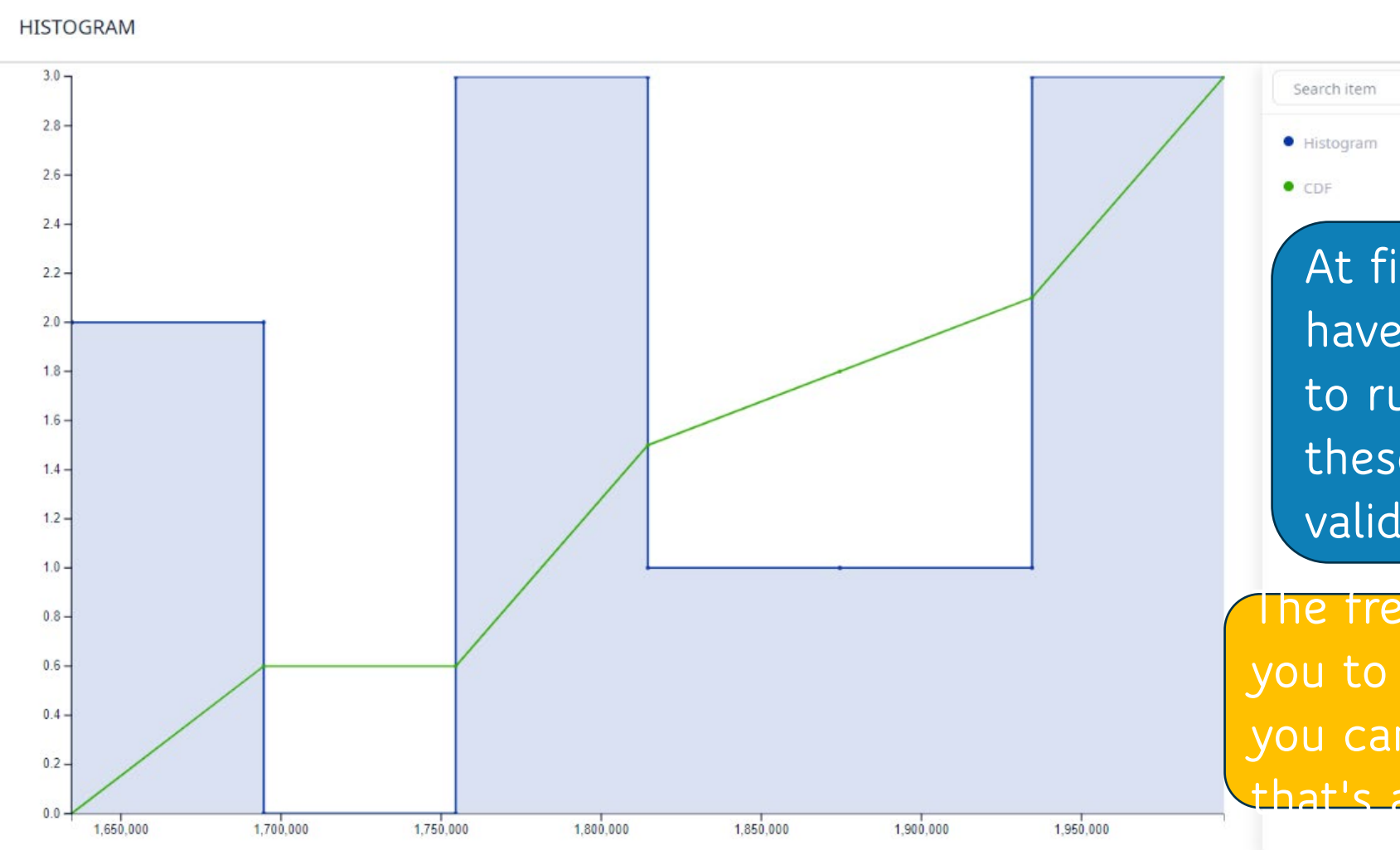


Man... At first glance, it seems that this is NOT so random. I'm going to take the AVERAGE as a reference value, as it seems significant and is "straddling" between the best and worst result.



Analyzing profit

- The histogram shows the RANGES of the most frequent values.



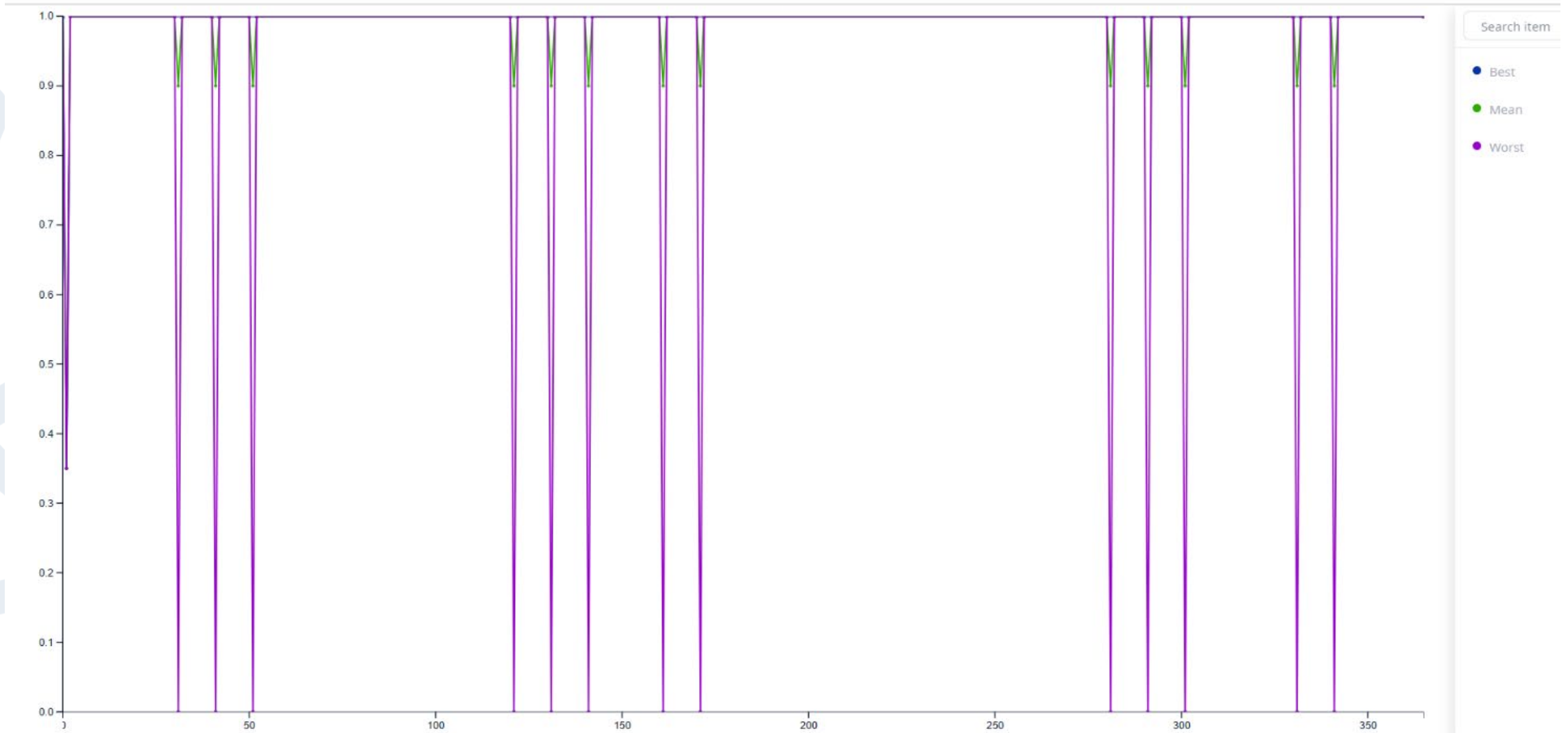
At first glance it seems that we have few values... we would have to run more iterations to see if these conclusions are definitely valid....

The free version of ALX only allows you to run a maximum of 10. What you can do is to run it many times, that's all



We look at the ELT for the B-M-W.

BEST-MEAN-WORST





Good job!

Sayonara baby!



Learning recap

1. Develop analytical and management skills to analyze the effect of random variations in the supply chain.
2. Develop risk analysis and management skills.
 1. To understand the relevance of randomized experiments.
 2. To be able to simulate probabilistic environments.